

App Dev Project Report

1. Student Details

Name: Kanchan

Roll Number: 24f1002360

Email: 24f1002360@ds.study.iitm.ac.in

About Me: I am a Student at IIT Madras BS Degree program with a deep interest in Software Development, web application development and data driven technologies. I enjoy building meaningful applications that combine learning, analytics, and user experience.

2. Project Details

Project Title: Trekking Management System

Problem Statement:

To design and build a web based application that allows users to **Admin, Trek Staff and Trekkers to manage trekking events**, such as trek bookings and staff supervising

Approach:

The application was developed using Flask with a modular architecture based on Blueprints. SQLAlchemy was used as the ORM with SQLite as the database. Bootstrap and Jinja2 templates were used to create a responsive user interface. The system follows role based authentication and authorization to provide different functionalities for Admin, Staff, and Trekker users while maintaining secure data management and proper booking workflows.

3. AI/LLM Declaration

AI tools such as **ChatGPT** were used for assistance in code structuring, debugging suggestions, and documentation formatting. The extent of AI/LLM usage is around **15-20%**, limited to **code suggestions and documentation formatting**.

The core logic, implementation, and integration of the application were developed manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5	Frontend styling and responsive design
Flask-Login	User authentication and session management
SQLite	Lightweight local database for storing user data
Werkzeug	Password hashing
HTML5 & CSS3	Web page structure and Styling

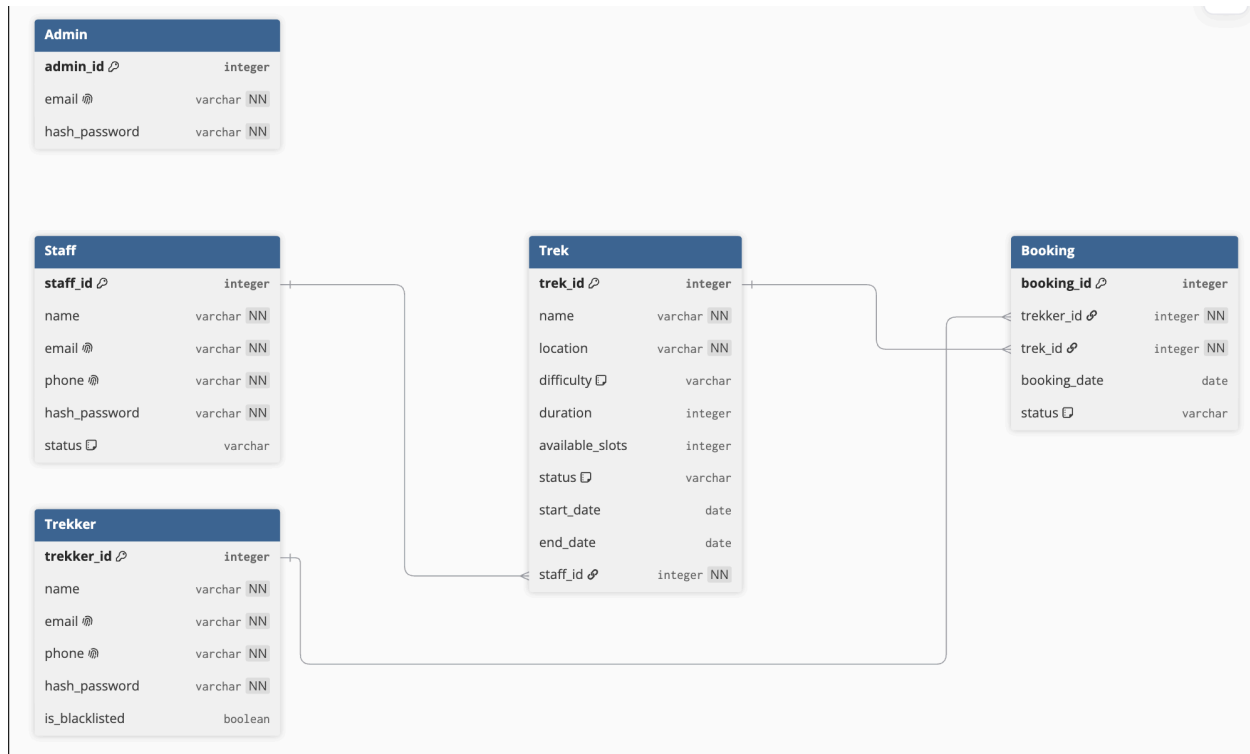
5. Database Schema / ER Diagram

Tables:

1. **Admin** - Stores administrator credentials. (admin_id, name, email, hash_password)
2. **Staff** - Stores trekking staff information. (staff_id, name, email, phone , hash_password, status)
3. **Trekker** - Stores registered trekkers. (trekker_id, name, email, phone, hash_password is_blacklisted)
4. **Trek** - Stores trekking event details. (trek_id, name, location, difficulty, duration, available_slots, status, start_date, end_date, staff_id)
5. **Booking** - Stores trek booking records. (booking_id, booking_date, status, trek_id, trekker_id)

Relationships:

- One-to-Many → Staff → Trek
- One-to-Many → Trek → Booking
- One-to-Many → Trekker → Booking



6. Architecture and Features

Architecture Overview

- **app.py** - main Flask application entry point
- **/models** - database models using SQLAlchemy
- **/routes** - Flask Blueprints for admin, staff and trekker routes
- **/templates** - Jinja2 HTML templates for UI
- **/static** - CSS for styling

Implemented Features

- Trekker, Staff registration and login along with dashboard
- Admin login and dashboard
- Admin can create and manage Treks
- Trekker/users can book available Treks and view their bookings
- Staff can manage Treks status
- Responsive UI built with Bootstrap
- Secure password hashing and user session handling

Additional Features

- Prevention of overbooking
 - Booking cancellation
 - Staff assignment and conflict checking
 - Search and filtering
 - Role based authorization
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7. Video Presentation

Drive Link:

https://drive.google.com/file/d/1QjTjZsuJZ7gbFnJIN43lbcwShLdc6_ya/view?usp=sharing
